



make My Day!

The '*make*' utility is one of the cleverest tools in the world of open source. It can be used in a variety of situations that are not just limited to programming. This article explores a little-known feature of '*make*'.

To begin with, here's the disclaimer: this article is not a tutorial or manual for *make*. Please read and understand the documentation for *make* before you try the experiment mentioned in this article. There are literally thousands of sources from where you can learn about *make*. The tutorial at www.wlug.org.nz/MakefileHowto is a neat and crisp one. The one at www.makelinux.net/make3/make3-PART-1.html is longer and more in depth.

With that out of our way, let's get started, shall we? A *make* rule is composed of the following:

target: prerequisites
 commands

A target is considered 'up to date' if it

exists and is newer than its prerequisites. The commands are executed only if the target is not up-to-date. The commands may end up updating the target. Each of the prerequisites can be the target in other *make* rules. Thus we have a whole tree of targets and prerequisites.

make is a very clever program. It works backwards, starting with the target of the first rule in the file. It walks down the recursion chain of target-prerequisite-target, until it finds a target that has no prerequisites, or whose prerequisites have no rules. Once it hits one of those, it walks back up its recursion chain and runs commands as necessary. It creates a recursion chain for every prerequisite it encounters that has a rule, and resolves it. It can also detect vicious circles of the following kind: A depends on B, B depends